Chapter 4

Animal Kingdom

Look around and you will observe different animals with different structures and forms. Over a million species have been described so far, which makes classification all the more important, and helps in assigning a systematic position to newly described species.

What the classification rests on

- in spite of differences in structure and form, there are fundamental features common to various individuals
- these concern the arrangement of cells, body symmetry, nature of coelom, and the patterns of the digestive, circulatory and reproductive systems

Levels of organisation

- all members of Animalia are multicellular, but they do not all show the same pattern of organisation of cells
- in sponges the cells are arranged as loose cell aggregates, the cellular level, with some division of labour among the cells
- in coelenterates the cells performing the same function are arranged into tissues, the tissue level
- in Platyhelminthes and higher phyla tissues are grouped into organs each specialised for a particular function, the organ level

In this chapter

4.1 Levels of organisation

4.3 Symmetry and germ layers

4.4 Coelom and notochord

4.5 The phyla, one by one

4.11 The vertebrate classes

4.12 Summary and exercises

Radial symmetry, any plane through the central axis gives two identical halves

Bilateral symmetry, identical left and right halves in one plane only

Figure 4.A (a) Radial symmetry (b) Bilateral symmetry

Diploblastic, an outer ectoderm and an inner endoderm

Mesoglea, the undifferentiated layer between them

Triploblastic, with mesoderm added between ectoderm and endoderm

Figure 4.B Showing germinal layers : (a) Diploblastic (b) Triploblastic

4.1 Levels of organisation, concluded

Organ system level

- in annelids, arthropods, molluscs, echinoderms and chordates the organs have associated into functional systems, each concerned with a specific physiological function
- organ systems in different groups show various patterns of complexity
- the digestive system in Platyhelminthes has a single opening serving as both mouth and anus, and is hence incomplete; a complete system has two openings, mouth and anus

Two kinds of circulatory system

- open type, in which blood is pumped out of the heart and the cells and tissues are bathed in it directly
- closed type, in which blood is circulated through a series of vessels of varying diameters, the arteries, veins and capillaries

4.2 Symmetry

Three body plans

- sponges are mostly asymmetrical, since no plane through the centre divides them into equal halves
- radial symmetry is where any plane passing through the central axis divides the organism into two identical halves, as in coelenterates, ctenophores and echinoderms
- bilateral symmetry is where the body divides into identical left and right halves in only one plane, as in annelids and arthropods

4.3 Diploblastic and triploblastic organisation

Two layers or three

- diploblastic animals have their cells arranged in two embryonic layers, an external ectoderm and an internal endoderm, as in coelenterates
- an undifferentiated layer, the mesoglea, lies between ectoderm and endoderm

Coelomate, body cavity lined by mesoderm

Pseudocoelomate, mesoderm in scattered pouches, cavity not lined

Acoelomate, no body cavity at all

Figure 4.C Diagrammatic sectional view of : (a) Coelomate (b) Pseudocoelomate (c) Acoelomate

Triploblastic animals

- those in which the developing embryo has a third germinal layer, the mesoderm, between ectoderm and endoderm
- this covers platyhelminthes through to chordates

4.4 Coelom, segmentation and notochord

Coelom

- the presence or absence of a cavity between the body wall and the gut wall matters greatly in classification
- a body cavity lined by mesoderm is a coelom, and animals possessing it are coelomates: annelids, molluscs, arthropods, echinoderms, hemichordates and chordates
- where the cavity is not lined by mesoderm, the mesoderm instead lying as scattered pouches between ectoderm and endoderm, the cavity is a pseudocoelom and the animals pseudocoelomates, as in aschelminthes
- animals in which the body cavity is absent are acoelomates, as in platyhelminthes

Segmentation

- in some animals the body is divided externally and internally into segments, with a serial repetition of at least some organs
- in earthworm this pattern is called metameric segmentation, and the phenomenon is metamerism

Notochord

- a mesodermally derived rod-like structure formed on the dorsal side during embryonic development in some animals
- animals with a notochord are chordates; those that do not form it are non-chordates, from porifera to echinoderms

Kingdom Animalia (multicellular)

Levels of Organisation	Symmetry	Body Cavity or Coelom	Phylum
Cellular level	mostly asymmetrical	acoelomata	Porifera
Tissue / Organ / Organ system	Radial	acoelomata	Coelenterata (Cnidaria), Ctenophora
Tissue / Organ / Organ system	Bilateral	Without body cavity (acoelomates)	Platyhelminthes
Tissue / Organ / Organ system	Bilateral	With false coelom (pseudocoelomates)	Aschelminthes
Tissue / Organ / Organ system	Bilateral	With true coelom (coelomates)	Annelida, Arthropoda, Mollusca, Echinodermata, Hemichordata, Chordata

Echinodermata exhibits radial or bilateral symmetry depending on the stage.

Figure 4.D Broad classification of Kingdom Animalia based on common fundamental features

Sycon, also called Scypha

Euspongia, the bath sponge

Spongilla, the fresh water sponge

Figure 4.E Examples of Porifera : (a) Sycon (b) Euspongia (c) Spongilla

4.5 Phylum Porifera

The sponges

- generally marine and mostly asymmetrical, these are primitive multicellular animals with cellular level of organisation
- they have a water transport or canal system: water enters through minute pores, the ostia, in the body wall into a central cavity, the spongocoel, and leaves through the osculum
- that pathway helps in food gathering, respiratory exchange and removal of waste
- choanocytes, or collar cells, line the spongocoel and the canals, and digestion is intracellular

Skeleton and reproduction

- the body is supported by a skeleton of spicules or spongin fibres
- sexes are not separate, that is they are hermaphrodite, with eggs and sperms produced by the same individual
- they reproduce asexually by fragmentation and sexually by formation of gametes
- fertilisation is internal and development indirect, with a larval stage morphologically distinct from the adult

Aurelia, the medusa form

Adamsia, the polyp form

Figure 4.F Examples of Coelenterata indicating outline of their body form : (a) Aurelia (Medusa) (b) Adamsia (Polyp)

Cnidoblast, the stinging cell of the tentacles and body

It holds the stinging capsule, the nematocyst

Figure 4.G Diagrammatic view of Cnidoblast

Sponges, concluded

- examples are Sycon, also called Scypha, Spongilla the fresh water sponge, and Euspongia the bath sponge

4.6 Phylum Coelenterata

Cnidoblasts give the phylum its other name

- aquatic, mostly marine, sessile or free-swimming and radially symmetrical
- the name cnidaria comes from the cnidoblasts or cnidocytes, which hold the stinging capsules or nematocysts, present on the tentacles and the body
- cnidoblasts are used for anchorage, defense and the capture of prey

Body plan

- they show tissue level of organisation and are diploblastic
- there is a central gastro-vascular cavity with a single opening, the mouth, on the hypostome, and digestion is both extracellular and intracellular
- some, such as corals, have a skeleton of calcium carbonate
- two basic body forms occur, the sessile cylindrical polyp such as Hydra and Adamsia, and the umbrella-shaped free-swimming medusa such as Aurelia, the jelly fish

Metagenesis, and examples

- cnidarians existing in both forms show alternation of generations, or metagenesis: polyps produce medusae asexually and medusae form polyps sexually, as in Obelia
- examples are Physalia the Portuguese man-of-war, Adamsia the sea anemone, Pennatula the sea-pen, Gorgonia the sea-fan and Meandrina the brain coral

Pleurobrachia, a sea walnut

Eight external rows of ciliated comb plates run along the body

Figure 4.H Example of Ctenophora (Pleurobrachia)

Tape worm, showing hooks and suckers

Liver fluke, a second parasitic flatworm

Figure 4.I Examples of Platyhelminthes : (a) Tape worm (b) Liver fluke

4.7 Phylum Ctenophora

Sea walnuts and comb jellies

- exclusively marine, radially symmetrical, diploblastic organisms with tissue level of organisation
- the body bears eight external rows of ciliated comb plates, which help in locomotion
- digestion is both extracellular and intracellular, and bioluminescence, the property of a living organism to emit light, is well-marked
- sexes are not separate, reproduction takes place only by sexual means, and fertilisation is external with indirect development; examples are Pleurobrachia and Ctenoplana

4.8 Phylum Platyhelminthes

The flatworms

- the body is dorso-ventrally flattened, hence the name
- they are mostly endoparasites found in animals including human beings
- bilaterally symmetrical, triploblastic and acoelomate, with organ level of organisation
- hooks and suckers are present in the parasitic forms, and some absorb nutrients from the host directly through the body surface

Excretion and reproduction

- specialised cells called flame cells help in osmoregulation and excretion
- sexes are not separate, fertilisation is internal, and development runs through many larval stages
- some members such as Planaria possess high regeneration capacity
- examples are Taenia the tapeworm and Fasciola the liver fluke

Roundworm, circular in cross-section

Male and female shown separately, since sexes are distinct

Figure 4.J Example of Aschelminthes: Roundworm

Nereis, an aquatic annelid with parapodia

Hirudinaria, the blood sucking leech

Figure 4.K Examples of Annelida : (a) Nereis (b) Hirudinaria

4.9 Phylum Aschelminthes

The roundworms

- the body is circular in cross-section, hence the name
- they may be free-living, aquatic and terrestrial, or parasitic in plants and animals
- organ-system level of organisation, bilaterally symmetrical, triploblastic and pseudocoelomate
- the alimentary canal is complete with a well-developed muscular pharynx, and an excretory tube removes body wastes from the body cavity through the excretory pore

Sexes and development

- sexes are separate, that is dioecious, and females are often longer than males
- fertilisation is internal, and development may be direct, the young resembling the adult, or indirect
- examples are Ascaris the roundworm, Wuchereria the filaria worm and Ancylostoma the hookworm

4.10 Phylum Annelida

Segmented worms

- aquatic, marine or fresh water, or terrestrial; free-living and sometimes parasitic
- organ-system level of organisation, bilateral symmetry, triploblastic, metamerically segmented and coelomate
- the body surface is distinctly marked into segments or metameres, whence the name Annelida, from the Latin annulus, little ring
- longitudinal and circular muscles help in locomotion, and aquatic annelids such as Nereis have lateral appendages, the parapodia, which help in swimming

Systems, and examples

- a closed circulatory system is present, and nephridia help in osmoregulation and excretion
- the neural system consists of paired ganglia connected by lateral nerves to a double ventral nerve cord
- Nereis is dioecious, but earthworms and leeches are monoecious; reproduction is sexual
- examples are Nereis, Pheretima the earthworm and Hirudinaria the blood sucking leech

Locust

Butterfly

Scorpion

Prawn

Figure 4.L Examples of Arthropoda : (a) Locust (b) Butterfly (c) Scorpion (d) Prawn

Pila, the apple snail, with its calcareous shell

Octopus, the devil fish

Figure 4.M Examples of Mollusca : (a) Pila (b) Octopus

4.11 Phylum Arthropoda

The largest phylum

- it includes the insects, and over two-thirds of all named species on earth are arthropods
- organ-system level of organisation, bilaterally symmetrical, triploblastic, segmented and coelomate
- the body is covered by a chitinous exoskeleton and consists of head, thorax and abdomen
- they have jointed appendages, from arthros meaning joint and poda meaning appendages

Systems and reproduction

- respiratory organs are gills, book gills, book lungs or the tracheal system, and the circulatory system is of open type
- sensory organs such as antennae, compound and simple eyes, and statocysts or balancing organs are present; excretion is through malpighian tubules
- they are mostly dioecious, fertilisation is usually internal, they are mostly oviparous, and development may be direct or indirect
- examples include Apis the honey bee, Bombyx the silkworm and Laccifer the lac insect; the vectors Anopheles, Culex and Aedes; Locusta the gregarious pest; and Limulus the king crab, a living fossil

4.12 Phylum Mollusca

The second largest phylum

- terrestrial or aquatic, marine or fresh water, with organ-system level of organisation; bilaterally symmetrical, triploblastic and coelomate
- the body is covered by a calcareous shell and is unsegmented, with a distinct head, muscular foot and visceral hump
- a soft spongy layer of skin forms a mantle over the visceral hump, and the space between hump and mantle is the mantle cavity, holding feather like gills with respiratory and excretory functions
- the anterior head region has sensory tentacles, and the mouth contains a file-like rasping organ for feeding, the radula

Asterias, the star fish

Ophiura, the brittle star

Figure 4.N Examples of Echinodermata : (a) Asterias (b) Ophiura

Proboscis at the anterior end

Collar behind the proboscis

Trunk, the long posterior part

Figure 4.O Balanoglossus

Molluscs, concluded

- they are usually dioecious and oviparous, with indirect development
- examples are Pila the apple snail, Pinctada the pearl oyster, Sepia the cuttlefish, Loligo the squid, Octopus the devil fish, Aplysia the sea-hare, Dentalium the tusk shell and Chaetopleura the chiton

4.13 Phylum Echinodermata

Spiny bodied animals

- they have an endoskeleton of calcareous ossicles, whence the name
- all are marine, with organ-system level of organisation; adults are radially symmetrical but larvae bilaterally symmetrical, and they are triploblastic and coelomate
- the digestive system is complete, with mouth on the lower ventral side and anus on the upper dorsal side
- the most distinctive feature is the water vascular system, which helps in locomotion, capture and transport of food, and respiration

Reproduction, and examples

- an excretory system is absent; sexes are separate and reproduction is sexual
- fertilisation is usually external and development indirect, with a free-swimming larva
- examples are Asterias the star fish, Echinus the sea urchin, Antedon the sea lily, Cucumaria the sea cucumber and Ophiura the brittle star

4.14 Phylum Hemichordata

Worm-like marine animals

- earlier considered a sub-phylum under Chordata, it is now placed as a separate phylum under non-chordata
- hemichordates have a rudimentary structure in the collar region called stomochord, similar to a notochord
- a small group with organ-system level of organisation; bilaterally symmetrical, triploblastic and coelomate, the body cylindrical with an anterior proboscis, a collar and a long trunk
- circulatory system is of open type, respiration is through gills, the excretory organ is the proboscis gland; sexes are separate, fertilisation external, development indirect; examples are Balanoglossus and Saccoglossus

Notochord running the length of the body

Nerve cord lying dorsal to it

Gill slits piercing the pharynx

Post-anal part behind the anus

Figure 4.P Chordata characteristics

Ascidia, a urochordate

Notochord present only in the larval tail

Figure 4.R Ascidia

4.15 Phylum Chordata

The three chordate characters

- a notochord, a dorsal hollow nerve cord and paired pharyngeal gill slits
- they are bilaterally symmetrical, triploblastic and coelomate, with organ-system level of organisation
- they possess a post anal tail and a closed circulatory system

Three subphyla

- Chordata divides into Urochordata or Tunicata, Cephalochordata and Vertebrata
- Urochordata and Cephalochordata are often called protochordates and are exclusively marine
- in Urochordata the notochord is present only in the larval tail, while in Cephalochordata it extends from head to tail and persists throughout life
- examples are Ascidia, Salpa and Doliolum among Urochordata, and Branchiostoma, the Amphioxus or Lancelet, among Cephalochordata

Vertebrata

- its members possess a notochord during the embryonic period, replaced in the adult by a cartilaginous or bony vertebral column
- so all vertebrates are chordates but all chordates are not vertebrates
- besides the basic chordate characters, vertebrates have a ventral muscular heart with two, three or four chambers, kidneys for excretion and osmoregulation, and paired appendages which may be fins or limbs

Table 4.A **Comparison of Chordates and Non-chordates**

S.No.	Chordates	Non-chordates
1.	Notochord present.	Notochord absent.
2.	Central nervous system is dorsal, hollow and single.	Central nervous system is ventral, solid and double.
3.	Pharynx perforated by gill slits.	Gill slits are absent.
4.	Heart is ventral.	Heart is dorsal (if present).
5.	A post-anal part (tail) is present.	Post-anal tail is absent.

Petromyzon, the lamprey

A sucking and circular mouth, without jaws

Figure 4.S A jawless vertebrate - Petromyzon

Scoliodon, the dog fish

Pristis, the saw fish

Figure 4.T Example of Cartilaginous fishes : (a) Scoliodon (b) Pristis

4.16 How Vertebrata divides

Two divisions, two super classes

- Agnatha lacks jaws and holds the class Cyclostomata
- Gnathostomata bears jaws and splits into the super classes Pisces, which bear fins, and Tetrapoda, which bear limbs
- Pisces holds Chondrichthyes and Osteichthyes; Tetrapoda holds Amphibia, Reptilia, Aves and Mammals

Class Cyclostomata

- all living members are ectoparasites on some fishes
- an elongated body bearing 6-15 pairs of gill slits for respiration, and a sucking circular mouth without jaws; the body is devoid of scales and paired fins
- cranium and vertebral column are cartilaginous, and circulation is of closed type
- marine, but they migrate to fresh water for spawning and die within a few days after it; their larvae return to the ocean after metamorphosis; examples are Petromyzon the lamprey and Myxine the hagfish

Class Chondrichthyes

- marine animals with streamlined body and cartilaginous endoskeleton; the mouth is located ventrally and the notochord persists throughout life
- gill slits are separate and without operculum, the gill cover
- the skin is tough with minute placoid scales, and the teeth are modified placoid scales directed backwards; the jaws are very powerful and these animals are predaceous
- lacking an air bladder, they have to swim constantly to avoid sinking

Hippocampus, the sea horse

Catla, a fresh water bony fish

Figure 4.U Examples of Bony fishes : (a) Hippocampus (b) Catla

Salamandra, a tailed amphibian

Rana, the frog

Figure 4.V Examples of Amphibia : (a) Salamandra (b) Rana

Chondrichthyes, concluded

- the heart is two-chambered, one auricle and one ventricle
- some have electric organs, as in Torpedo, and some a poison sting, as in Trygon
- they are cold-blooded, or poikilothermous, lacking the capacity to regulate body temperature
- sexes are separate, in males the pelvic fins bear claspers, fertilisation is internal and many are viviparous; examples are Scoliodon, Pristis, Carcharodon and Trygon

4.17 Class Osteichthyes

The bony fishes

- marine and fresh water fishes with bony endoskeleton and streamlined body; the mouth is mostly terminal
- four pairs of gills covered by an operculum on each side, and skin covered with cycloid or ctenoid scales
- an air bladder is present, which regulates buoyancy, and the heart is two-chambered
- cold-blooded, sexes separate, fertilisation usually external, mostly oviparous with direct development; examples are Exocoetus, Hippocampus, Labeo, Catla, Clarias, Betta and Pterophyllum

4.18 Class Amphibia

Dual life

- from the Greek amphi meaning dual and bios meaning life, they can live in aquatic and terrestrial habitats
- most have two pairs of limbs; the body divides into head and trunk, and a tail may be present in some
- the skin is moist and without scales, the eyes have eyelids, and a tympanum represents the ear
- alimentary canal, urinary and reproductive tracts open into a common chamber, the cloaca, which opens to the exterior

Respiration and reproduction

- respiration is by gills, lungs and through the skin
- the heart is three-chambered, two auricles and one ventricle, and they are cold-blooded
- sexes are separate, fertilisation is external, they are oviparous and development is indirect
- examples are Bufo, Rana, Hyla, Salamandra and Ichthyophis, the limbless amphibia

Chameleon, the tree lizard

Crocodilus, the crocodile

Chelone, the turtle

Naja, the cobra

Figure 4.W Reptiles: (a) Chameleon (b) Crocodilus (c) Chelone (d) Naja

4.19 Class Reptilia

Creeping and crawling

- the class name refers to their mode of locomotion, from the Latin repere or reptum, to creep or crawl
- mostly terrestrial, the body covered by dry and cornified skin with epidermal scales or scutes
- they have no external ear openings, and a tympanum represents the ear; limbs, when present, are two pairs

Heart, temperature and reproduction

- the heart is usually three-chambered, but four-chambered in crocodiles
- reptiles are poikilotherms, and snakes and lizards shed their scales as skin cast
- sexes are separate, fertilisation is internal, they are oviparous and development is direct
- examples are Chelone, Testudo, Chameleon, Calotes, Crocodilus, Alligator and Hemidactylus, with the poisonous snakes Naja, Bangarus and Vipera

4.20 Class Aves

Feathers and flight

- the characteristic features are feathers, and most can fly except flightless birds such as the ostrich; they possess a beak
- the forelimbs are modified into wings, while the hind limbs generally have scales and are modified for walking, swimming or clasping tree branches
- the skin is dry and without glands, except the oil gland at the base of the tail
- the endoskeleton is fully ossified and the long bones are hollow with air cavities, that is pneumatic

Digestion, heart and warmth

- the digestive tract has additional chambers, the crop and gizzard
- the heart is completely four-chambered
- they are warm-blooded, or homoiothermous, able to maintain a constant body temperature

Neophron, the vulture

Struthio, the ostrich

Psittacula, the parrot

Pavo, the peacock

Figure 4.X Some birds : (a) Neophron (b) Struthio (c) Psittacula (d) Pavo

Ornithorhynchus, the platypus

Macropus, the kangaroo

Pteropus, the flying fox

Balaenoptera, the blue whale

Figure 4.Y Some mammals : (a) Ornithorhynchus (b) Macropus (c) Pteropus (d) Balaenoptera

Birds, concluded

- respiration is by lungs, and air sacs connected to the lungs supplement it
- sexes are separate, fertilisation is internal, they are oviparous and development is direct
- examples are Corvus, Columba, Psittacula, Struthio, Pavo, Aptenodytes and Neophron

4.21 Class Mammalia

Habitat and limbs

- found in a variety of habitats: polar ice caps, deserts, mountains, forests, grasslands and dark caves, and some have adapted to fly or live in water
- the most unique mammalian characteristic is the presence of milk producing mammary glands, by which the young are nourished
- they have two pairs of limbs, adapted for walking, running, climbing, burrowing, swimming or flying

The mammalian body

- the skin is unique in possessing hair, and external ears or pinnae are present
- different types of teeth are present in the jaw, and the heart is four-chambered
- they are homoiothermous and respiration is by lungs
- sexes are separate, fertilisation is internal, they are viviparous with few exceptions and development is direct

Examples

- oviparous: Ornithorhynchus, the platypus
- viviparous: Macropus, Pteropus, Camelus, Macaca, Rattus, Canis, Felis, Elephas, Equus, Delphinus, Balaenoptera, Panthera tigris and Panthera leo

Table 4.B Salient Features of Different Phyla in the Animal Kingdom

Phylum	Level of Organisation	Symmetry	Coelom	Segmentation	Digestive System	Circulatory System	Respiratory System	Distinctive Features
Porifera	Cellular	Various	Absent	Absent	Absent	Absent	Absent	Body with pores and canals in walls.
Coelenterata (Cnidaria)	Tissue	Radial	Absent	Absent	Incomplete	Absent	Absent	Cnidoblasts present.
Ctenophora	Tissue	Radial	Absent	Absent	Incomplete	Absent	Absent	Comb plates for locomotion.
Platyhelminthes	Organ and organ-system	Bilateral	Absent	Absent	Incomplete	Absent	Absent	Flat body, suckers.
Aschelminthes	Organ-system	Bilateral	Pseudocoelomate	Absent	Complete	Absent	Absent	Often worm-shaped, elongated.
Annelida	Organ-system	Bilateral	Coelomate	Present	Complete	Present	Absent	Body segmentation like rings.
Arthropoda	Organ-system	Bilateral	Coelomate	Present	Complete	Present	Present	Exoskeleton of cuticle, jointed appendages.
Mollusca	Organ-system	Bilateral	Coelomate	Absent	Complete	Present	Present	External skeleton of shell usually present.
Echinodermata	Organ-system	Radial	Coelomate	Absent	Complete	Present	Present	Water vascular system, radial symmetry.
Hemichordata	Organ-system	Bilateral	Coelomate	Absent	Complete	Present	Present	Worm-like with proboscis, collar and trunk.
Chordata	Organ-system	Bilateral	Coelomate	Present	Complete	Present	Present	Notochord, dorsal hollow nerve cord, gill slits with limbs or fins.

4.22 Summary

What lets us classify at all

- level of organisation, symmetry, cell organisation, coelom, segmentation and notochord have enabled the broad classification of the animal kingdom
- besides these fundamental features there are many other distinctive characters specific to each phylum or class

The non-chordate phyla

- Porifera are multicellular animals of cellular level of organisation with characteristic flagellated choanocytes
- coelenterates have tentacles and bear cnidoblasts, and are mostly aquatic, sessile or free-floating; ctenophores are marine with comb plates
- platyhelminths have a flat body and bilateral symmetry, the parasitic forms showing distinct suckers and hooks; aschelminthes are pseudocoelomates and include parasitic and non-parasitic roundworms

From annelids to hemichordates

- annelids are metamerically segmented animals with a true coelom
- arthropods are the most abundant group, marked by jointed appendages
- molluscs have a soft body surrounded by an external calcareous shell, the body covered with external skeleton made of chitin
- echinoderms possess a spiny skin, their most distinctive feature being the water vascular system; hemichordates are a small group of worm-like marine animals with a cylindrical body of proboscis, collar and trunk

Chordates

- Chordata includes animals possessing a notochord either throughout life or during early embryonic life
- other common features are the dorsal hollow nerve cord and paired pharyngeal gill slits
- some vertebrates lack jaws, the Agnatha, while most possess them, the Gnathostomata; Agnatha is represented by Cyclostomata, the most primitive chordates, ectoparasites on fishes
- Gnathostomata has two super classes, Pisces and Tetrapoda; Chondrichthyes and Osteichthyes bear fins and are grouped under Pisces, the Chondrichthyes being marine fishes with cartilaginous endoskeleton

The tetrapod classes, concluded

- Amphibia, Reptilia, Aves and Mammalia have two pairs of limbs and are grouped under Tetrapoda
- amphibians have adapted to live both on land and in water; reptiles are marked by dry and cornified skin, and limbs are absent in snakes
- fishes, amphibians and reptiles are poikilothermous, that is cold-blooded
- Aves are warm-blooded with feathers and forelimbs modified into wings, the hind limbs adapted for walking, swimming, perching or clasping; mammals are unique in mammary glands and hairs on the skin, and commonly exhibit viviparity

4.23 Exercises

From this chapter

1 What difficulties would you face in classifying animals if common fundamental features were not taken into account?
2 Given a specimen, what steps would you follow to classify it?
3 How useful is the study of the nature of body cavity and coelom in the classification of animals?
4 Distinguish between intracellular and extracellular digestion.
5 What is the difference between direct and indirect development?
6 What peculiar features do you find in parasitic platyhelminthes?
7 What reasons can you think of for the arthropods constituting the largest group of the animal kingdom?
8 Water vascular system is characteristic of which of these groups: (a) Porifera (b) Ctenophora (c) Echinodermata (d) Chordata
9 All vertebrates are chordates but all chordates are not vertebrates. Justify the statement.
10 How important is the presence of an air bladder in Pisces?
11 What modifications are observed in birds that help them fly?
12 Could the number of eggs or young ones produced by an oviparous and a viviparous mother be equal? Why?
13 Segmentation in the body is first observed in which of these: (a) Platyhelminthes (b) Aschelminthes (c) Annelida (d) Arthropoda

From this chapter, continued

1. Match these against one another: (a) Operculum (b) Parapodia (c) Scales (d) Comb plates (e) Radula (f) Hairs (g) Choanocytes (h) Gill slits, and (i) Ctenophora (ii) Mollusca (iii) Porifera (iv) Reptilia (v) Annelida (vi) Cyclostomata and Chondrichthyes (vii) Mammalia (viii) Osteichthyes
2. Prepare a list of some animals that are found parasitic on human beings.